اسلایدهای آموزشی

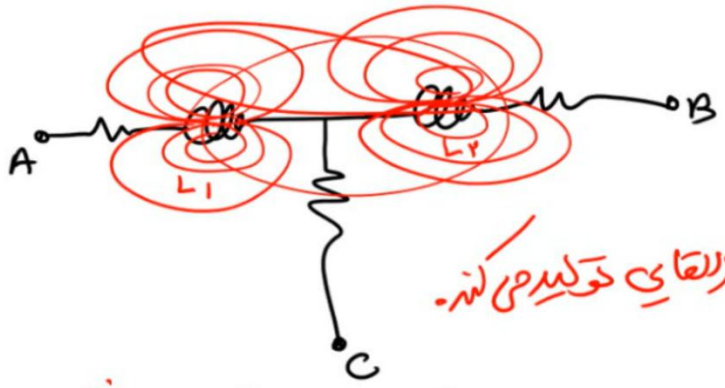
سیستم های انرژی الکتریکی ۱

دکتر مهران حاجی آقاپور

استادیار دانشکده مهندسی برق
دانشگاه شهید بهشتی

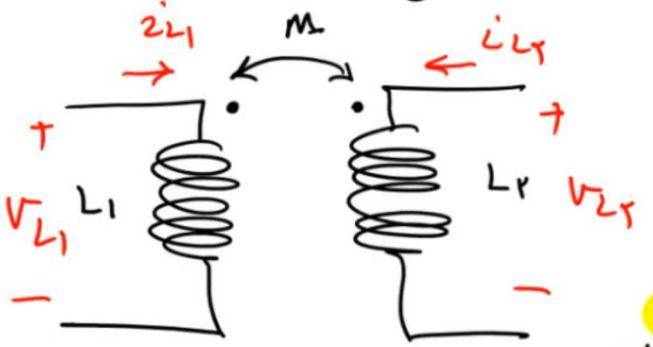
مدارات تزویج

مدارات تزویج



قانون فاراده:
تغییرات شار الکتریکی، ولتاژ القایی تولید می کند.

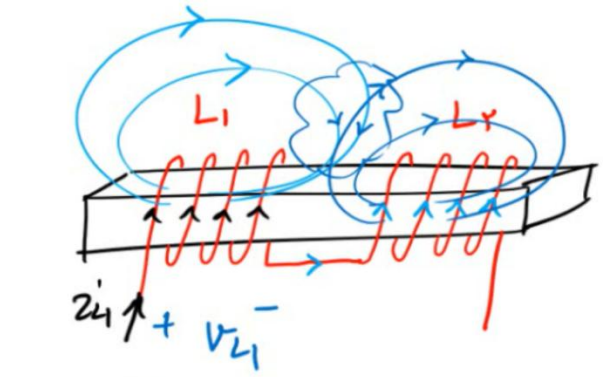
M : القاکتانس متقابل (H)



$$\begin{cases} v_{L_1} = L_1 \frac{d}{dt} i_{L_1} \pm M \frac{d i_{L_2}}{dt} \\ v_{L_2} = L_2 \frac{d}{dt} i_{L_2} \pm M \frac{d i_{L_1}}{dt} \end{cases}$$

+ : شار سلف‌ها هم جهت باشد.
− : شار سلف‌ها در خلاف هم باشد.

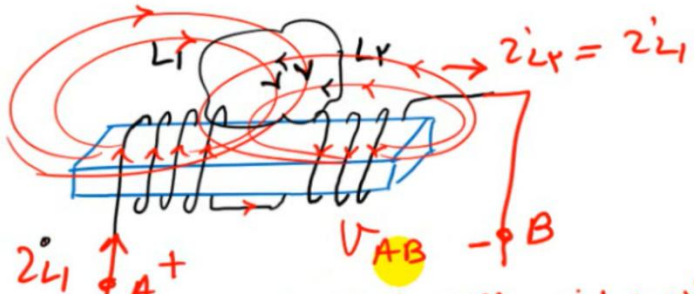
مدارات تزویج



خودمغناطیس

$$v_{L_1} = L_1 \frac{d}{dt} 2i_1 - M \frac{d}{dt} 2i_1$$

$$v_{L_1} = (L_1 - M) \frac{d}{dt} 2i_1$$



$$v_{L_1} = L_1 \frac{d}{dt} 2i_1 + M \frac{d 2i_1}{dt}$$

$$= (L_1 + M) \frac{d 2i_1}{dt}$$

بجای سلف اصلی توان سلف $L_1 + M$ قرار داد.

مدارات تزویج

$$V_{AB} = v_{L_1} + v_{L_2}$$

$$= (L_1 + M)\frac{d}{dt} i_{L_1} + (L_2 \frac{d}{dt} i_{L_2} + M \frac{d}{dt} i_{L_2})$$

$$(L_1 + L_2 + 2M)\frac{d}{dt} i_{L_1}$$

حالت دوم



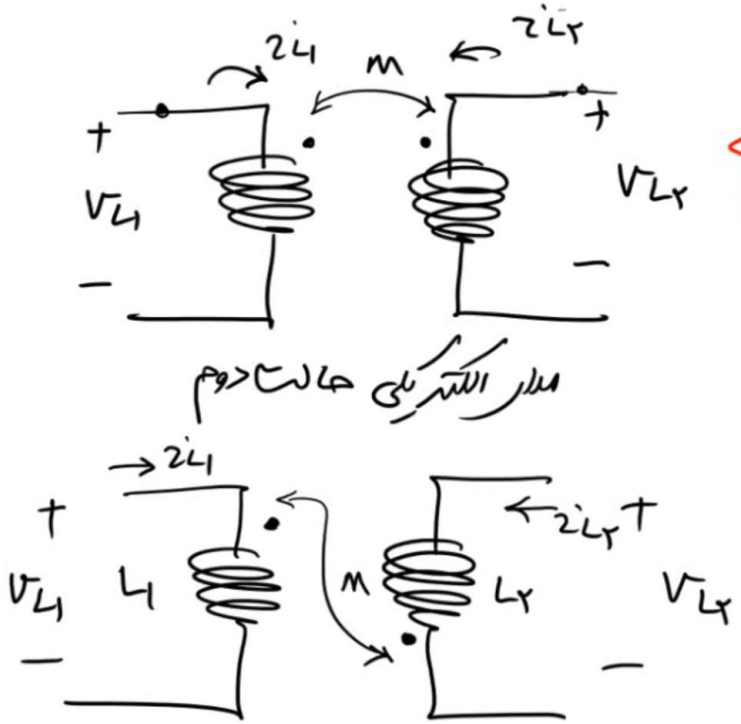
$$L_{eq} = L_1 + L_2 + 2M$$

حالت اول



$$L_{eq} = L_1 + L_2 - 2M$$

مدارات تزویج



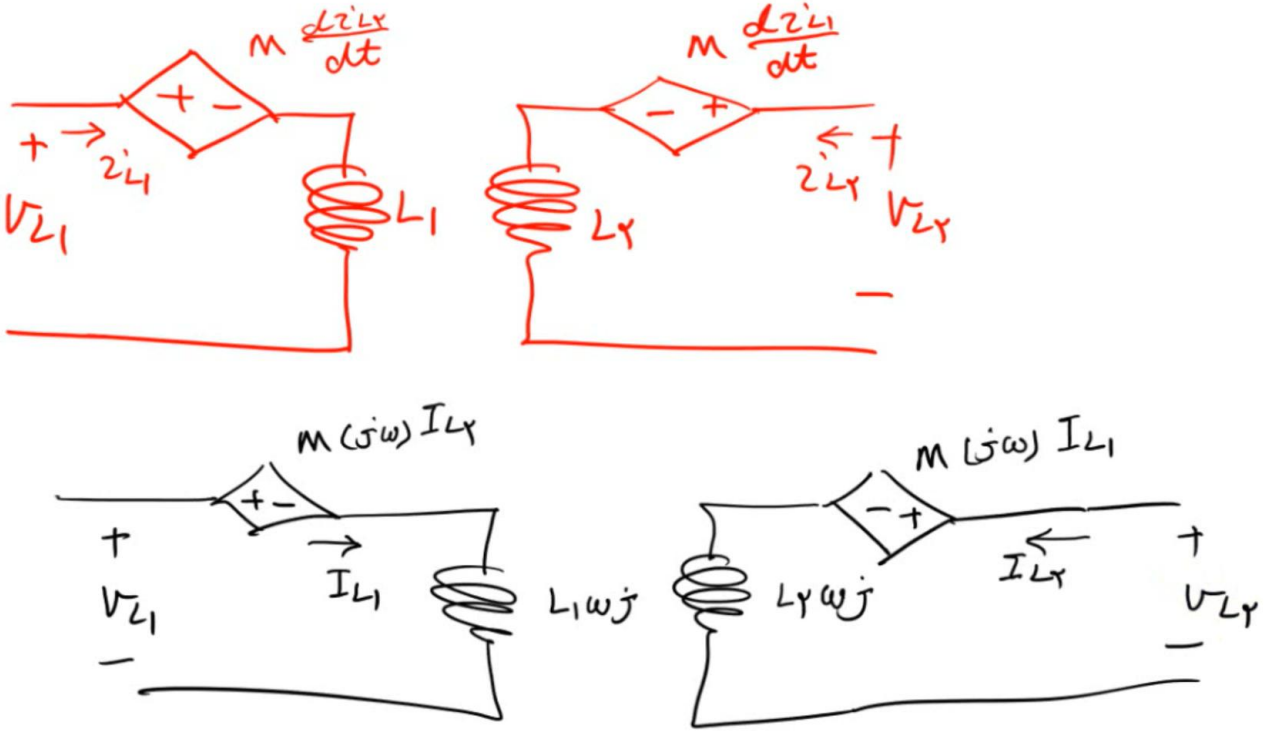
$$\begin{cases} v_{L1} = L_1 \frac{d}{dt} i_1 + M \frac{d}{dt} i_2 \\ v_{L2} = L_2 \frac{d}{dt} i_2 + M \frac{d}{dt} i_1 \end{cases}$$

مثبت

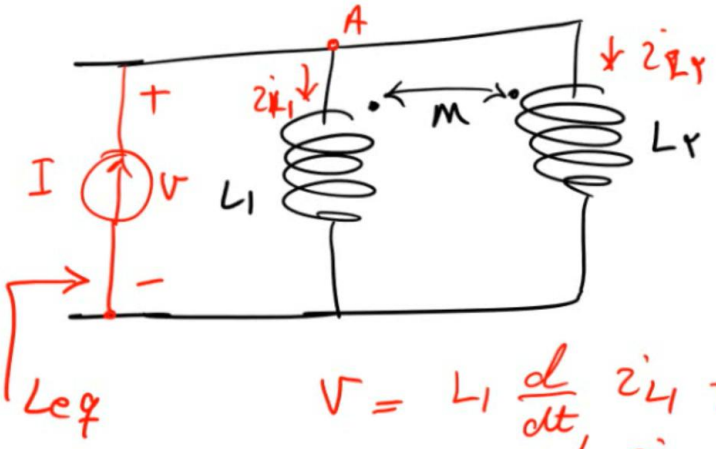
$$\begin{cases} v_{L1} = L_1 \frac{d}{dt} i_1 - M \frac{d}{dt} i_2 \\ v_{L2} = L_2 \frac{d}{dt} i_2 - M \frac{d}{dt} i_1 \end{cases}$$

منفی

مدارات تزویج



مدارات تزویج



$$I = i_{L_1} + i_{L_2}$$

$$V = L_1 \frac{d}{dt} i_{L_1} + M \frac{d}{dt} i_{L_2}$$

$$= L_2 \frac{d}{dt} i_{L_2} + M \frac{d}{dt} i_{L_1}$$

$$(L_1 I_{L_1})\omega + (M I_{L_2})\omega = (L_2 I_{L_2})\omega + (M I_{L_1})\omega$$

$$(L_1 - M) I_{L_1} = (L_2 - M) I_{L_2} \Rightarrow \frac{I_{L_2}}{I_{L_1}} = \frac{L_1 - M}{L_2 - M}$$

مدارات تزویج

$$I = I_{L_1} + \frac{L_1 - M}{L_2 - M} I_{L_1}$$

$$I_{L_1} = \frac{I}{1 + \frac{L_1 - M}{L_2 - M}} = \frac{L_2 - M}{L_1 + L_2 - 2M} I$$

$$V = (L_1 I_{L_1} + M I_{L_2}) \omega j$$

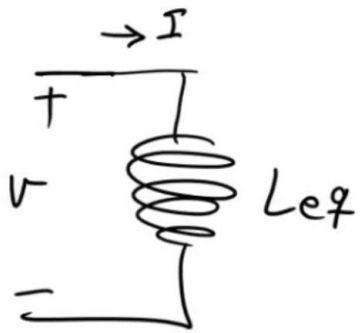
$$V = \left(L_1 \times \frac{L_2 - M}{L_1 + L_2 - 2M} \times I + M \times \frac{L_1 - M}{L_2 - M} \times \frac{L_2 - M}{L_1 + L_2 - 2M} I\right) \times \omega j$$

$$V = \left(\frac{L_1 L_2 - M L_1 + M L_1 - M^2}{L_1 + L_2 - 2M}\right) (\omega j) I$$

مدارات تزویج

$$V = \left(\frac{L_1 L_2 - M^2}{L_1 + L_2 - M^2}\right) \omega j I$$

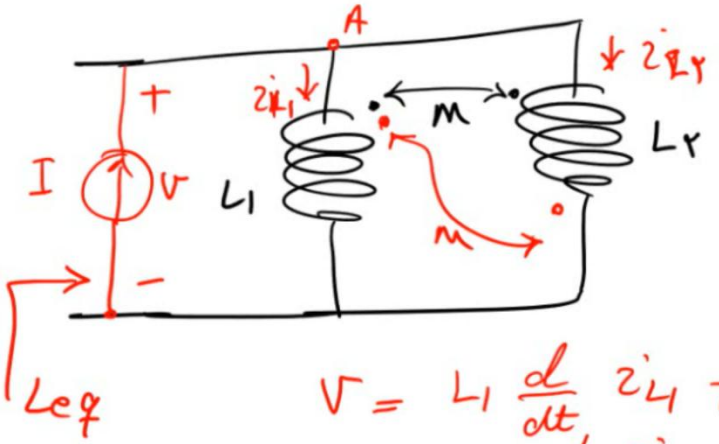
$$V = L_{eq} (\omega j) I$$



$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - M^2} \geq 0$$

ضریب کوپلاژ $= k = \frac{M}{\sqrt{L_1 L_2}}$ $\quad 0 < k \leq 1$

مدارات تزویج



$I = i_{L_1} + i_{L_2}$ ✓

$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$

القاکتانس مثبت : + ، القاکتانس منفی : − }

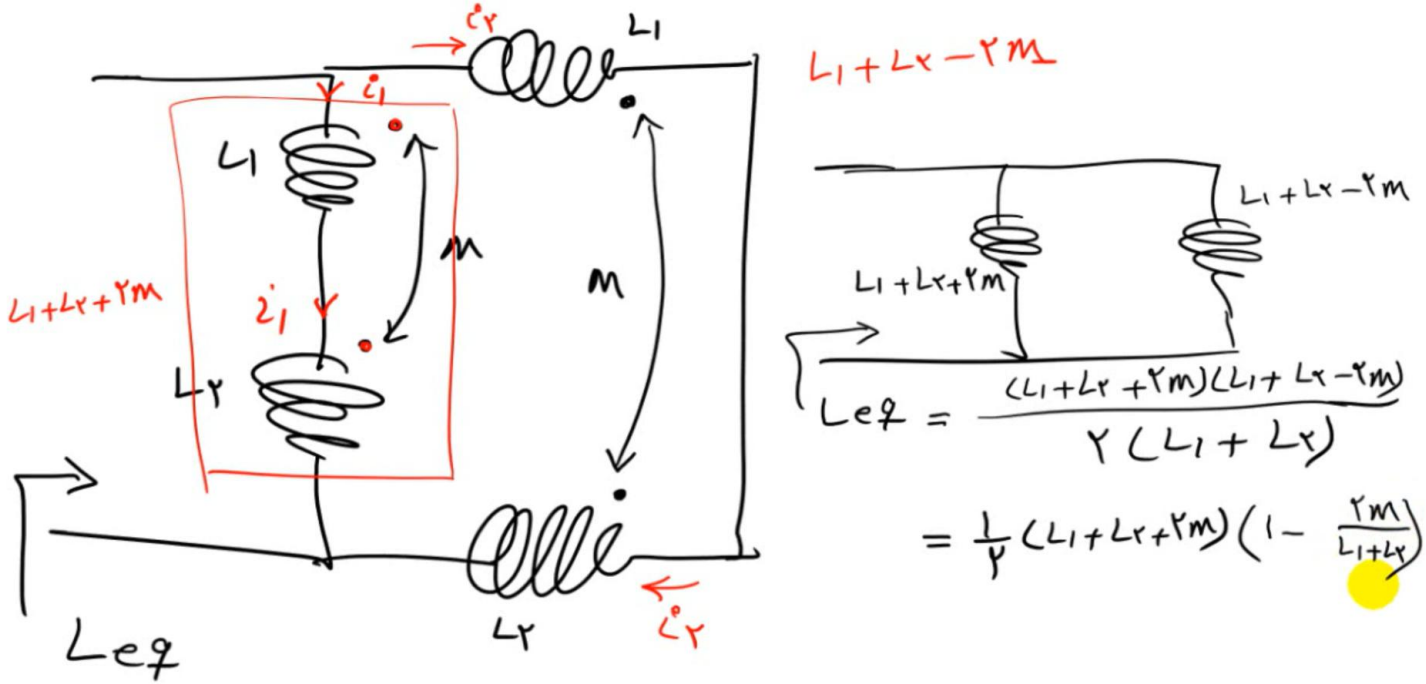
$V = L_1 \frac{d}{dt} i_{L_1} + M \frac{d}{dt} i_{L_2}$ ←

$= L_2 \frac{d}{dt} i_{L_2} + M \frac{d}{dt} i_{L_1}$

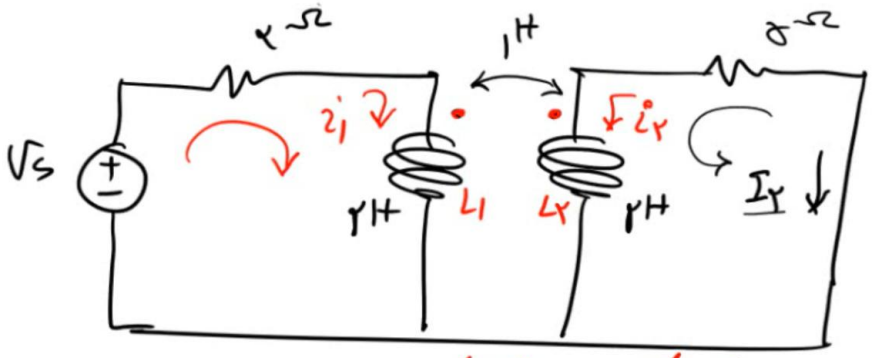
$(L_1 I_{L_1})\omega j + (M I_{L_2})\omega j = (L_2 I_{L_2})\omega j + (M I_{L_1})\omega j$

$(L_1 - M) I_{L_1} = (L_2 - M) I_{L_2} \Rightarrow \frac{I_{L_2}}{I_{L_1}} = \frac{L_1 - M}{L_2 - M}$

مدارات تزویج



مدارات تزویج



$$\begin{cases} v_{L_1} = 2\frac{d}{dt} i_1 + \frac{d}{dt} i_2 \\ v_{L_2} = 2\frac{d}{dt} i_2 + \frac{d}{dt} i_1 \end{cases}$$

$$\rightarrow \begin{cases} V_s = 2i_1 + 2D\, i_1 + D\, i_2 \\ 5 i_2 + 2D\, i_2 + D\, i_1 = 0 \end{cases} \rightsquigarrow i_1 = \frac{(2D+5) i_2}{-D}$$

مدارات تزویج

$$\begin{cases} V_s = (2+2D)\left(\frac{2D+3}{-D}\right) i_2 + D\, i_2 \\ I_1 = -i_2 \end{cases}$$

$$V_s = (2+2D)\left(\frac{2D+3}{D}\right) I_1 - D\, I_1$$

$$D\,V_s = 2(D+1)(2D+3)\, I_1 - D^2 I_1$$

$$= 2(2D^2+5D+3)\, I_1 - D^2 I_1$$

$$= (3D^2+10D+6)\, I_1$$

$$\longrightarrow \frac{d}{dt} V_s(t) = 3\frac{d^2}{dt^2} I_1(t) + 10\frac{d}{dt} I_1 + 6\, I_1(t)$$

مدارات تزویج

$$v_s = 10 \sin t$$

حوزه : sin $\qquad v_s = 10 \qquad \omega = 1$

$$(j\omega)\, V_s = 4(j\omega)^2 I_2 + 14(j\omega) I_2 + 10 I_2$$

$$I_2 = \frac{10j}{-4 + 14j + 10} = \frac{10j}{7 + 14j}$$

$$I_2 = \frac{10}{7} \cdot \frac{j}{1 + 2j} = \frac{10}{7} \cdot \frac{j(1 - 2j)}{1 + 4} = \frac{2}{7}(2 + j)$$

$$I_2 = \frac{2\sqrt{5}}{7} \tan^{-1}\left(\frac{1}{2}\right) \qquad i_2(t) = \frac{2\sqrt{5}}{7} \sin\left(t + \tan^{-1}\left(\frac{1}{2}\right)\right)$$